

Pretentious Self-Parsing and Self-Evaluation of λ -Calculus

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String “ $(\lambda. 0)$ ”

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Concrete and Abstract Syntax

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Abstract Syntax

$t ::= \text{Var}(n)$	(Variable)
$\text{Lam}(t)$	(Function)
$\text{App}(t, t)$	(Application)

(De Bruijn index: $\text{Var}(0)$ = Closest variable, $\text{Var}(1)$ = Next closest, ...)
e.g.) $\lambda x. \lambda y. x y = \lambda. \lambda. 10$

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- 1 Represent each token
- 2 Represent each AST constructor
- 3 Parser `e_parse` from token list to AST
- 4 Evaluator `e_eval` from AST to λ -expression

Theory: Algebraic Data Types and Scott Encoding

- Algebraic data type (ADT, 조립식 타입): Sum, product, and recursive types

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Constructor		Scott Encoding
$\llbracket \text{LPAR} \rrbracket$	=	$\lambda l r a v. l$
$\llbracket \text{RPAR} \rrbracket$	=	$\lambda l r a v. r$
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Case statement

$$t \ fv \ fl \ fa = \begin{cases} fv \ n & (t = \text{Var}(n)) \\ fl \ t' & (t = \text{Lam}(t')) \\ fa \ t_1 \ t_2 & (t = \text{App}(t_1, t_2)) \end{cases}$$

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Example: $\lambda x. \lambda y. x = \lambda. \lambda. 1$

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- Scott: Top-level v, l, a operate on the immediate subexpressions directly (Top-down)
- For parsing and evaluating, **Scott encoding** is more suitable!

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`( $\lambda n. \text{lookup } \text{lst } n$ )`

(Var: Lookup)

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- Implementation in GitHub repository!

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